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# EDA Bench: An Execution-Based Benchmark for Language Model Agents on Functional PCB Construction

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## Abstract

Language-model agents increasingly operate electronic design automation tools: they search datasheets, select components, edit KiCad projects, run design checks, and iterate after failures. Reliable PCB construction requires a realized board to preserve the target design’s external electrical behavior. EDA Bench is an execution-based benchmark with 35 KiCad PCB reconstruction tasks derived from public open-hardware projects, spanning connector breakouts, power-path circuits, microcontroller systems, motor drivers, camera and compute adapters, and high-density carriers. Each harness receives a prompt and a standard KiCad environment and produces a complete KiCad project. The behavioral evaluator parses the PCB, extracts routed copper, checks declared external ports, models trace parasitics and short faults, runs ngspice simulations, and applies KiCad ERC and DRC checks. External-I/O behavior determines the score, while reference, failure, and mutation controls calibrate it: all 35 frozen references score 1.0, and all 35 structural-failure controls score at most 0.15. The strongest evaluated web-enabled harness scores 4.58% overall and builds 21 of 35 projects. These results establish complete, electrically correct PCB reconstruction as an extremely difficult task for current agent systems.

## 1 Introduction

Language-model agents can now operate development tools over long horizons. Execution-based benchmarks measure the resulting artifacts in the environments where they run: an agent modifies an artifact, executes tools, observes failures, and iterates. SWE-bench evaluates agents on real GitHub issues whose solutions must pass repository tests [Jimenez et al., 2023]. OSWorld extends execution-based evaluation to open-ended desktop and web tasks in real computer environments [Xie et al., 2024]. ProgramBench evaluates reconstruction from an executable and documentation, using behavioral tests to compare the rebuilt program with the reference [Yang et al., 2026]. Together, these benchmarks establish behavior in an execution environment as a strong measure of artifact quality.

Hardware design has an analogous workflow. Engineers search for parts, inspect datasheets, draft schematics, route boards, run ERC and DRC checks, simulate where possible, and produce manufacturing files. EDA evaluation must cover routed connectivity, electrical behavior, manufacturability constraints, and external interfaces. Existing CAD and EDA benchmarks address text-to-CAD generation [Khan et al., 2024], geometric CAD validation [Epoch AI, 2026, CAD Arena, 2026, Barkley et al., 2026], schematic generation [Al Hasan et al., 2026, Zou et al., 2026, Luo et al., 2026], schematic understanding [Lu et al., 2026], board-level schematic design [Qiu et al., 2026], hardware bug repair [Cui et al., 2026], and PCB placement and routing reasoning [Li et al., 2026]. EDA Bench extends this landscape to complete routed KiCad projects whose external electrical behavior is checked directly.

Table 1: EDA Bench evaluation pipeline. Project parsing, electrical checks, and simulation score each complete harness output.

| Stage             | Artifact or operation                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Task input        | The harness receives a fixed prompt, runtime task pack, tool environment, model configuration, and access policy.                   |
| Harness execution | The agent operates in the Dockerized KiCad environment and writes a project to <code>/workspace/final_project</code> .              |
| Project parsing   | The grader locates KiCad files and extracts connectors, pads, nets, board outline, and routed copper geometry.                      |
| Contract binding  | External ports from the submitted board are matched to task-declared I/O contracts and expected measurements.                       |
| Execution checks  | The grader applies KiCad ERC and DRC diagnostics, ngspice simulations, routed-I/O checks, parasitic models, and short-fault checks. |
| Score reporting   | The evaluator returns a normalized task score, hard-cap diagnostics, build status, and run artifacts.                               |

EDA Bench evaluates complete agent harnesses on source-backed KiCad PCB reconstruction tasks. A harness receives a fixed task prompt and runtime assets, then writes a complete KiCad project in `/workspace/final_project`. KiCad provides an open-source EDA environment for schematic capture, integrated simulation, PCB layout, 3D rendering, and manufacturing export [KiCad Project, 2026]. The tasks are based on public hardware projects, and the default harnesses have web access. This setting combines auditable source designs with the realistic source-discovery workflow used by tool-operating agents.

The scoring system measures observable behavior at the realized board’s external electrical interfaces. It parses, checks, and simulates each project against a task contract, with ngspice as the open-source circuit-simulation backend [ngspice contributors, 2026]. Different internal implementations can earn the same score when they expose the required behavior. Missing, shorted, swapped, isolated, or weakly driven external nets reduce the score through direct electrical-interface checks. Frozen references, structural-failure controls, mutation controls, and representation-invariance checks establish the evaluator’s sensitivity to engineering-relevant PCB changes.

## 2 Benchmark overview

EDA Bench evaluates PCB reconstruction as an execution task. For each task, a harness receives a fixed prompt, runtime task pack, tool environment, model configuration, and access policy, then produces a complete KiCad project in `/workspace/final_project`. The automated evaluator parses the KiCad files, extracts external connectors and routed copper, applies task-declared stimuli and probes, runs simulations, checks ERC and DRC diagnostics, and assigns a normalized score based on the realized board behavior.

Table 1 summarizes the execution and grading pipeline. Each released task is derived from a public hardware project and has a frozen reference design. The evaluated runtime pack contains the prompt and task-visible assets, while the grading pack contains the reference project, source snapshot, controls, contracts, and evaluator implementation. This separation yields auditable tasks with a clear boundary between agent-visible inputs and grading materials. The public-source access policy measures PCB reconstruction with source discovery and tool use available to the evaluated harness.

Scores characterize complete evaluated harnesses. A harness may include prompting, web search, source discovery, KiCad tool use, repair loops, and project-generation logic. The protocol evaluates the system architecture as a whole, beyond just a standalone model, because practical EDA automation depends on operating the full workflow.

The score measures declared external-I/O behavior under the released task contracts. Equivalent internal implementations receive credit when they realize the required behavior, while missing, isolated, shorted, swapped, unrouted, or weakly driven external nets trigger corresponding deductions. This execution-based scope complements fabrication, regulatory, thermal, EMC, and vendor-silicon review.

Table 2: Task distribution and representative categories in EDA Bench.

| Difficulty | Tasks | Representative task families                                     |
|------------|-------|------------------------------------------------------------------|
| Very easy  | 2     | Compact connector breakouts and small adapters                   |
| Easy       | 5     | I/O breakouts, logic-level conversion, USB/UART basics           |
| Medium     | 4     | Power-path boards, serial transceivers, small controller boards  |
| Hard       | 12    | Compute adapters, sensor/camera boards, multi-connector carriers |
| Very hard  | 7     | Dense mixed-signal boards and larger interface carriers          |
| Extreme    | 5     | Large baseboards and high-density compute boards                 |

### 3 Source-backed tasks

EDA Bench contains 35 source-backed tasks. Each task is anchored to a public upstream hardware project and a frozen reference design. The task set covers six difficulty levels, shown in Table 2.

Each released task requires a source design, a frozen KiCad reference, separate runtime and grading packs, an external-I/O contract, control submissions, license triage, and evaluator support. The resulting 35-task set prioritizes deeply checked board reconstructions across six difficulty levels.

The task catalog includes compact breakouts and converters such as `usb_c_female_breakout`, `mcp23017_breakout`, and `rs485_transceiver_breakout`. It also includes power and control boards such as `bq24295_power_path_board`, `picopd`, and `robotont_driver_board`, compute and camera or display adapters such as `cm4_csi_adapter`, `m2_pcie_adapter`, and `ov9281_dual_camera_board`, and large carrier boards such as `cm4_baseboard`, `jetson_orin_baseboard`, and `zynq_som`.

Each task has two artifact forms. The runtime task pack is mounted for the evaluated agent and contains the prompt, task metadata, and runtime assets. The public grading pack contains the reference project, upstream snapshot, control submissions, license notes, `io_contract.json`, `functional_contract.json`, and local tests. Separate packs preserve a clean task boundary and support exact reproduction of grading results.

### 4 Harness protocol

A harness is evaluated as a complete system. It receives the task prompt and must produce a KiCad project in `/workspace/final_project`. Built-in harnesses run inside Docker with KiCad, ngspice, Python, Node, internet access, and the configured model interface. The evaluated harnesses are Codex CLI with GPT-5.5 web search at low, medium, high, and xhigh reasoning, Pi with Gemini 3.1 Pro Preview web high, and Pi with DeepSeek V4 Pro web high.

This protocol evaluates the system architecture as a whole, beyond just a standalone model. A one-shot generator, a tool-using agent, and a multi-agent harness can all target the same task interface. EDA work is iterative: the system creates files, runs KiCad tools, inspects failures, searches for upstream sources, and repairs invalid output.

Each reported run includes task prompts, command lines, usage summaries, grader outputs, final projects, and generated artifacts. These records allow final projects to be regraded without rerunning model harnesses.

### 5 Scoring system

The scorer evaluates a submitted KiCad project through explicit I/O simulation. Unsupported, missing, or malformed submissions receive zero score. Supported tasks declare their external ports, stimuli, probes, simulator models, expected measurements, and tolerances. For each submission, the evaluator locates and parses the KiCad project, runs ERC and DRC diagnostics, extracts connectors, pads, nets, and routed copper geometry, binds submitted external ports to the task contract, constructs task-declared probes and stimuli, computes task-family measurements and hard caps, and returns a normalized score with diagnostic artifacts.

Table 3: Representative task-family checks layered on the generic external-I/O evaluator. These checks bind measured signals to task contracts using deterministic circuit and channel models.

| Task family                             | Declared behavior                                                                                     | Common hard caps                                                      |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Connector breakouts and simple adapters | Connector pins and power/ground pads, continuity, shorts, routed coverage, pin/net semantic agreement | Swapped pins, split nets, isolated pads, power-ground shorts          |
| Serial and level-shifting interfaces    | USB/UART/RS-485/logic-side boundary ports, signal transfer families, active-device realization        | Missing active path, weak external mapping, missing routed I/O        |
| Power-path and charger boards           | Input, battery/load, configuration, and status ports, power path presence, external safety            | Missing active power role, shorts, missing component-function profile |
| Camera/display/compute carriers         | High-density connector families and differential pairs, routed families, pair plausibility            | Missing high-speed pair families, weak routing, shorts, split nets    |
| Large mixed-signal baseboards           | Multi-rail power, GPIO, debug, serial, high-speed, and peripheral interfaces                          | Incomplete connector semantics, missing active realization, shorts    |

The primary score is a normalized task score in  $[0, 1]$ . A benchmark aggregate is a per-task difficulty-weighted mean:

$$S = \frac{\sum_{i=1}^N w(d_i) s_i}{\sum_{i=1}^N w(d_i)}, \quad w(d_i) \in \{1.0, 1.5, 2.0, 3.0, 4.0, 5.0\}. \quad (1)$$

We also report the unweighted mean over tasks. Build success means that the submission produced a KiCad project that the grader could parse and evaluate. It is not equivalent to functional success.

The main score combines three observable properties of the submitted PCB: external connector semantics, routed-copper behavior, and task-required active and manufacturing plausibility. The evaluator applies deterministic ngspice stimuli at declared external ports, samples boundary waveforms, and caps scores for missing routed copper, swapped external pins, shorts, missing active circuitry, unpowered active devices, invalid high-speed pair families, and gross fabrication implausibility. Reference designators, component inventory, footprint similarity, outline similarity, and route-shape similarity provide supporting diagnostics.

Within each task, declared measurements produce normalized sub-scores for external-port mapping, routed continuity, simulation behavior, and task-family realization checks. These sub-scores are combined according to the task contract to form a soft functional score. Hard caps then impose upper bounds for structural failures that make the board externally incorrect or physically implausible, such as shorts, missing routed I/O, swapped connector pins, or absent active circuitry. The final task score is the resulting capped functional score. This design allows partial credit for partially correct external behavior while preventing submissions with severe structural faults from receiving high scores through unrelated checks.

## 6 Artifact validation

Because EDA Bench uses task-specific evaluation, the release includes controls that calibrate score interpretation. The frozen reference project serves as the positive control and must score 1.0. The structural-failure control removes active internal circuitry when available or removes the external boundary on passive boards; it must score below the 0.75 release threshold.

**Evaluator validity.** Score changes should track engineering-relevant changes to declared external-I/O behavior. Positive controls establish that human-designed reference boards are realizable and score 1.0. Structural-failure controls measure sensitivity to function-breaking changes. Invariance

Table 4: Evaluator-validity evidence. The controls calibrate the released score as an external-I/O contract measure.

| Evidence                                                      | What it supports                                                                                                           |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 35/35 released frozen human-designed reference PCBs score 1.0 | The tasks are solvable, and real upstream designs satisfy their declared contracts.                                        |
| 35/35 released structural fail canaries score at most 0.15    | The score is not reducible to file existence, KiCad syntax, or component-shell matching.                                   |
| 162 mutation canaries pass validation                         | The evaluator is sensitive to shorts, unrouted nets, isolated pads, missing outlines, and missing high-speed-like routing. |
| Refdes-renamed USB-C board scores 1.0                         | Renaming connector reference designators preserves the score.                                                              |
| Plausible USB-C route-width edit remains at least 0.85        | Plausible route-width changes preserve high scores.                                                                        |
| Tiny-trace USB-C control scores at most 0.35                  | Manufacturability caps affect scores for physically implausible routing.                                                   |

Table 5: Complete web-enabled harness baselines. The primary score is difficulty-weighted, and the unweighted score is the mean task score.

| Harness                            | Weighted | Unweighted | Builds |
|------------------------------------|----------|------------|--------|
| Codex GPT-5.5 web low              | 0.0124   | 0.0143     | 31/35  |
| Codex GPT-5.5 web medium           | 0.0279   | 0.0314     | 35/35  |
| Codex GPT-5.5 web high             | 0.0293   | 0.0229     | 11/35  |
| Codex GPT-5.5 web xhigh            | 0.0458   | 0.0500     | 21/35  |
| Pi Gemini 3.1 Pro Preview web high | 0.0113   | 0.0086     | 1/35   |
| Pi DeepSeek V4 Pro web high        | 0.0000   | 0.0000     | 0/35   |

controls confirm that benign representation changes, such as reference-designator renaming or plausible route-width edits, have little effect.

A full local sweep on May 6, 2026 verified all 35 released task canaries: every pass canary scored 1.0, and every fail canary scored at most 0.15. The generated-mutation sweep graded 162 mutation canaries across released task families. All tasks passed, with every reference scoring 1.0 and every structural fail canary scoring at most 0.15. Mutations covered missing board outlines, unrouted external nets, isolated external pads, power or signal pads tied to ground, and missing high-speed-like routes for high-speed carrier tasks.

Together, these controls show that the score accepts human-designed reference boards, rejects common function-breaking PCB mutations, and remains stable under several superficial implementation changes.

## 7 Experiments

We evaluate six web-enabled harness configurations on the 35 released tasks. All runs use the same task prompts, task packs, Docker image, and grading protocol. Table 5 reports a May 6, 2026 no-model regrade of saved final projects under the evaluator revision used for this manuscript.

Complete PCB reconstruction remains extremely difficult. The Codex GPT-5.5 web medium harness builds all 35 released projects and scores 2.79%, while the xhigh harness builds 21 projects and reaches the best score of 4.58%. The large separation between build rate and behavioral score shows that parseable KiCad output captures only an early stage of the task. Reasoning and tool-use choices also change the distribution of electrical failures.

Difficulty breakdowns further indicate that the benchmark is not saturated by easy tasks. The Codex GPT-5.5 web xhigh harness reaches 13.75% on medium tasks and 7.08% on hard tasks, but only 2.00% on extreme tasks. The gap grows as boards require larger external contracts, active circuitry, dense routing, or high-speed interfaces.

**Statistical interpretation.** The result table contains one complete released-task sweep per configuration. It supports task-level failure analysis, absolute difficulty estimates, and comparisons between

Table 6: Completed no-web calibration runs. These rows are reported separately from the default web leaderboard because access policy is an evaluated condition.

| Harness                          | Weighted | Unweighted | Builds |
|----------------------------------|----------|------------|--------|
| Codex GPT-5.5 no-web medium      | 0.0000   | 0.0000     | 0/35   |
| Codex GPT-5.5 no-web high        | 0.0000   | 0.0000     | 0/35   |
| Codex GPT-5.5 no-web xhigh       | 0.0000   | 0.0000     | 0/35   |
| Codex GPT-5.4 no-web xhigh       | 0.0000   | 0.0000     | 0/35   |
| Codex GPT-5.4-mini no-web medium | 0.0000   | 0.0000     | 0/35   |
| Codex GPT-5.4-mini no-web high   | 0.0000   | 0.0000     | 0/35   |

build rate and behavioral score. Repeated independent sweeps would be required for pairwise model-ranking significance tests.

### 7.1 Access-policy calibration

EDA Bench reports access policy as part of the evaluated system. Six no-web Codex calibration sweeps use the same released tasks and grader revision as the web-enabled runs with web access disabled. All six scored zero and produced no buildable projects, showing that the default prompts and built-in Codex harnesses combine public-source discovery with KiCad operation. The separate table isolates this access-policy effect from the default web-enabled leaderboard.

## 8 Failure analysis

The best evaluated configuration scores 4.58% overall, placing complete PCB reconstruction among the hardest evaluated agent tasks. One harness produces buildable KiCad projects for all 35 tasks yet scores 2.79%, showing that most difficulty lies beyond file generation. Codex GPT-5.5 configurations with different reasoning settings also produce distinct build and score tradeoffs, reinforcing the importance of evaluating the complete system architecture.

Saved grading artifacts expose the main failure modes. Among buildable Codex GPT-5.5 web xhigh submissions, the most common caps were split required external-net continuity, isolated external-I/O pads, wrong external-I/O pin/net mapping, missing or weak component-function realization, missing or implausible high-speed pair geometry, and weak active-device power integrity. The all-building Codex GPT-5.5 web medium run shows a similar pattern: wrong external mapping, missing component-function realization, split required external nets, same-layer trace shorts, and isolated external-I/O pads.

These caps correspond to concrete electrical failures. Submissions often instantiate connector footprints with plausible labels while leaving required I/O pads isolated, splitting required nets across disconnected copper, swapping external pin semantics, omitting active circuitry, or routing high-speed-like interfaces implausibly. The behavioral score measures these difficult board-level requirements directly.

## 9 Related work

**Execution-based software-agent benchmarks.** Execution-based software benchmarks evaluate agents by running the artifacts they produce. SWE-bench frames software engineering as repository-level issue resolution, where an agent edits a real codebase and must pass tests associated with the target issue [Jimenez et al., 2023]. OSWorld evaluates multimodal agents in real desktop and web environments with custom execution-based evaluation scripts [Xie et al., 2024]. ProgramBench studies a more reconstructive setting in which agents rebuild software from an executable and documentation, and behavioral tests are generated to compare the candidate program with the reference executable [Yang et al., 2026]. EDA Bench adopts this behavioral-evaluation perspective but applies it to PCB reconstruction, where correctness depends on realized geometry, external connectivity, and electrical-interface behavior rather than software tests alone.

**CAD generation and geometric evaluation.** Text2CAD introduced a large-scale text-to-parametric-CAD dataset and model for generating CAD command sequences from natural-language prompts [Khan et al., 2024]. CadEval and CAD Arena provide evaluation settings for text-to-CAD models using geometric validity, rendering, and prompt-level comparisons [Epoch AI, 2026, CAD Arena, 2026]. CADSmith further emphasizes programmatic geometric validation for generated CadQuery models [Barkley et al., 2026]. These works motivate execution-based evaluation for engineering artifacts, but their primary correctness signals are geometric rather than electrical. PCB reconstruction requires connectivity, routing, short avoidance, active-circuit realization, and external-I/O behavior in addition to shape or layout plausibility.

**Circuit, schematic, and PCB benchmarks.** Recent EDA benchmarks study several adjacent tasks. CircuitLM and PCBSchemaGen generate or synthesize circuit schematics from natural-language prompts and use structural or constraint-based checks to improve schematic validity [Al Hasan et al., 2026, Zou et al., 2026]. SchGen studies PCB schematic generation using semantic-grounded code representations [Luo et al., 2026]. OmniSch evaluates multimodal schematic understanding and schematic-to-graph reasoning over real-world diagrams [Lu et al., 2026]. HWE-Bench evaluates board-level schematic design with datasheet context and hardware bug repair over repository-scale hardware projects [Qiu et al., 2026, Cui et al., 2026]. PCB-Bench evaluates LLM reasoning over placement, routing, and PCB design comprehension using text, images, and complete PCB projects [Li et al., 2026]. EDA Bench differs by requiring an agent to produce a complete routed KiCad PCB project and by scoring the realized board through external-I/O simulation and PCB-geometry checks rather than schematic-only validity, visual reasoning, or placement/routing question answering.

## 10 Limitations

The released tasks use public sources, and evaluated web-enabled harnesses may discover upstream designs. Scores therefore measure EDA task solving under fixed prompts, tools, task-pack revisions, run records, and access policies. The separate no-web calibration quantifies the effect of source access for the built-in harnesses.

The 35 released tasks emphasize depth: every task requires a real source design, a reference project, task contracts, controls, license review, and grader support. Additional source-backed packs and a sealed private split would expand coverage across more PCB families.

The behavioral evaluator uses deterministic generic models for external-I/O behavior. Manufacturing review adds electrical safety, regulatory, high-speed protocol, thermal, power-sequencing, and production-readiness analysis. Richer per-task SPICE, IBIS, S-parameter, and field-solver models would extend fidelity beyond the four currently calibrated board families.

Task artifacts inherit mixed upstream hardware licenses. The public artifact includes only score-supported tasks whose runtime and grading packs can be redistributed under the release policy.

## 11 Conclusion

EDA Bench evaluates complete agent systems on electrically grounded PCB reconstruction. Real public hardware projects, explicit I/O simulation, routed-copper analysis, and KiCad checks turn external-interface behavior into a reproducible score. Current systems can often produce parseable KiCad projects, while the best evaluated harness reaches only 4.58% on the full board-level task. This large gap establishes reliable PCB reconstruction as an extremely difficult target for agentic EDA systems.

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## A Artifact availability

The source repository contains the runner and harness code, task-pack loader and grader support, public documentation, package tests, and the task-pack manifest. Python wheels and source distributions exclude concrete task instances, reference projects, upstream snapshots, and control submissions. The Hugging Face task dataset contains runtime task packs, full grader task packs, prompts, references, upstream snapshots, control submissions, task metadata, Croissant metadata, and license notes.



The public task dataset is hosted at <https://huggingface.co/datasets/eda-bench-neurips-2026/eda-bench-tasks>. The results use task-pack revision c64b9169d6d8c0d73614e14c7050cf036aeb1559. The public code release is hosted at <https://github.com/CAD-bench/eda-bench>.

Run records include task prompts, command lines, usage summaries, grader outputs, final projects, and generated artifacts. Storage-light campaigns retain final reports, final projects, grading artifacts, and byte-count summaries while omitting full Docker image uploads and raw model transcripts.

## B Validation controls

Table 7: Validation controls for score interpretation.

| Control                           | Observed result                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Frozen reference project          | 35/35 released references score 1.0                                                                          |
| Structural fail canary            | 35/35 released fail canaries score at most 0.15, below the 0.75 release threshold                            |
| Generated mutation canaries       | 162 generated mutation canaries across released task families pass validation                                |
| Calibrated task-family checks     | USB-C, RS-485, BQ24295, and M.2/PCIe references score 1.0 and their mutation canaries remain below threshold |
| Refdes-invariant equivalent board | USB-C reference with renamed connector refdes scores 1.0                                                     |
| Missing routed copper             | USB-C and NanoUPDI route-removal controls score 0.0                                                          |
| Plausible route-width edit        | USB-C route-width edit remains at least 0.85                                                                 |
| Implausibly tiny traces           | Manufacturability cap applies and USB-C tiny-trace control scores at most 0.35                               |
| Missing or unparsable project     | Returns 0.0 and build failure                                                                                |

## C Failure-mode diagnostics

Table 8: Common electrical failure modes exposed by the evaluator in otherwise buildable submissions.

| Failure mode                       | Observable symptom                                                                                                              | Why it affects score                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Split required external nets       | Pins that should belong to the same external interface are assigned to disconnected copper regions or separate nets             | Boundary stimuli cannot propagate through the declared interface, even when connector footprints are present |
| Isolated external I/O pads         | Required pads are placed but not connected to routed copper or active circuitry                                                 | The board exposes a physical pad without realizing the electrical behavior required by the contract          |
| Wrong pin or net mapping           | External connector pins are swapped, mislabeled, or attached to semantically incorrect nets                                     | The board may appear visually plausible while driving or observing the wrong external signal                 |
| Missing active realization         | Components required for level shifting, power-path behavior, transceiver behavior, or control logic are absent or not connected | Passive connectivity alone cannot implement the task-required circuit behavior                               |
| Weak active-device power integrity | Active devices are present but lack required power, ground, or enable connectivity                                              | The simulated or inferred behavior cannot be attributed to a realizable powered circuit                      |
| Implausible high-speed geometry    | Differential or high-speed-like interface families are missing, weakly routed, or routed with implausible geometry              | Dense carrier and compute tasks require routed interface families with plausible physical realization        |

## D Baseline run records

The May 6, 2026 result table regrades saved final projects without rerunning model harnesses. Source runs were: `codex__gpt-5.5-web-low_20260505T200208Z_2719115`, `codex__gpt-5.5-web-medium_20260506T010715Z_2812175`, `codex__gpt-5.5-web-medium_20260506T010715Z_2812182`, `codex__gpt-5.5-web-medium_20260506T032737Z_2884928`, `codex__gpt-5.5-web-medium_20260506T010716Z_2812190`, `codex__gpt-5.5-web-high_20260505T200208Z_2719113`, `codex__gpt-5.5-web-xhigh_20260506T032737Z_2884940`, `codex__gpt-5.5-web-xhigh_20260506T032737Z_2884938`, `codex__gpt-5.5-web-xhigh_20260506T032737Z_2884943`, `codex__gpt-5.5-web-xhigh_20260506T034126Z_2891808`, `pi__gemini-3.1-pro-preview-web-high_20260505T184850Z_2697190`, and `pi__deepseek__deepseek-v4-pro-web-high_20260505T200208Z_2719116`.

The release’s regrade entry point accepts one or more source-run identifiers, uses the released task manifest, downloads the saved `submission/final_project` directories, and runs the local evaluator. Passing the four Codex GPT-5.5 web medium identifiers above reproduces that aggregate row.

Completed no-web calibration runs are present under these run paths: `evals/harnesses/codex__gpt-5.5-no_web-medium_20260507T033319Z_3432012`, `evals/harnesses/codex__gpt-5.5-no_web-high_20260507T033319Z_3432018`, `evals/harnesses/codex__gpt-5.5-no_web-xhigh_20260507T033319Z_3432017`, `evals/harnesses/codex__gpt-5.4-no_web-xhigh_20260507T033528Z_3439753`, `evals/harnesses/codex__gpt-5.4-mini-no_web-medium_20260507T033724Z_3448333`, and `evals/harnesses/codex__gpt-5.4-mini-no_web-high_20260507T033724Z_3448341`.

## E Cost and compute accounting

The completed GPT-5.5 web runs used storage-light run records. The combined Codex GPT-5.5 low/medium/high/xhigh campaign uploaded about 185.6 MiB. Across all six final harnesses, the total upload was about 193.1 MiB.

Table 9: Recorded Codex GPT-5.5 web campaign accounting.

| Harness                  | Estimated OpenAI API USD | Estimated Codex credits | HF records |
|--------------------------|--------------------------|-------------------------|------------|
| Codex GPT-5.5 web low    | 38.22                    | 955.41                  | 48.9 MB    |
| Codex GPT-5.5 web medium | 84.40                    | 2110.05                 | 49.5 MB    |
| Codex GPT-5.5 web high   | 49.39                    | 1234.64                 | 27.6 MB    |
| Codex GPT-5.5 web xhigh  | 182.75                   | 4568.64                 | 68.6 MB    |

The estimates are computed from recorded token usage, current provider pricing, and the Codex token-based rate card. They make leaderboard costs and storage requirements directly auditable.

## F License and redistribution

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